

MANIPAL ACADEMY OF HIGHER EDUCATION
MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
Deep Learning [IT_4442]

Marks: 50
mins.

Duration: 180

PART-A

Answer all the questions.

1. a. Compare the bias-variance tradeoff in machine learning. How do you balance the two when tuning a model to achieve better generalization? (5)

- Definition of Bias and Variance:
 - Bias: Error introduced by approximating a real-world problem with a simplified model. High bias leads to underfitting.
 - Variance: Error introduced by the model's sensitivity to small fluctuations in the training set. High variance leads to overfitting.
- Tradeoff:
 - A model with high bias and low variance underfits, while one with high variance and low bias overfits.
 - Balancing: Use techniques like cross-validation, early stopping, and hyperparameter tuning (e.g., regularization, tree depth) to achieve a balance between bias and variance, ensuring the model generalizes well.

b. Analyze the differences between classification and regression in supervised learning. How do the techniques used in each differ in terms of model evaluation? (5)

- Classification:
 - Predicts discrete labels or categories.
 - Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix.
- Regression:
 - Predicts continuous values.

- Evaluation Metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), R-Squared, Root Mean Squared Error (RMSE).
- Key Difference:
 - Classification uses categorical evaluation metrics, while regression uses continuous error metrics.

2. a. Analyze the components of a deep feedforward network. How do the input, hidden, and output layers work together to learn from data? (5)

- Components:
 - Input Layer: Takes the raw input data (features) and passes it to the first hidden layer.
 - Hidden Layers: Process the data using weights, biases, and activation functions. The number of layers and neurons determines the depth of the network.
 - Output Layer: Produces the final prediction based on the activations from the last hidden layer.
- Learning Process:
 - The layers work together by applying transformations and learning abstract features through successive layers, refining the data representation until the output is close to the target value.

b. Apply the backpropagation algorithm to adjust weights in a neural network. How does the gradient of the loss function guide the training process? (5)

- Backpropagation:
 - Involves computing the gradient of the loss function with respect to the weights, which is done by applying the chain rule in reverse.
 - Gradient Descent: The gradient tells how much the weights should be adjusted to minimize the loss function. A negative gradient indicates weights should be increased, while a positive gradient indicates they should be decreased.
 - Training Process: The network iteratively adjusts weights by propagating errors backward from the output to the input layers.

3. a. Apply data augmentation techniques to improve the generalization of a deep learning model. How does augmentation help in preventing overfitting on limited datasets? (5)

- Data Augmentation:
 - Techniques include rotating, flipping, cropping, and scaling images or adding noise to data.
 - Prevents Overfitting: By artificially enlarging the training set, augmentation helps the model generalize better by learning from more diverse examples, thus reducing the risk of overfitting to a limited dataset.

b. Apply regularization techniques such as L1 and L2 to prevent overfitting. How do they affect the complexity of the model and its performance? (5)

- L1 Regularization (Lasso):
 - Adds a penalty equal to the absolute value of weights, promoting sparsity (some weights become zero).
 - Effect: Encourages simpler models by reducing irrelevant features.
- L2 Regularization (Ridge):
 - Adds a penalty equal to the square of the weights, preventing large weights and reducing model complexity.
 - Effect: Helps in preventing overfitting by shrinking the model's weights.

4. a. Evaluate the importance of choosing an appropriate learning rate for neural network optimization. How does the learning rate schedule affect model training and convergence? (5)

- Learning Rate Importance:
 - A too-high learning rate can cause the model to overshoot the minimum, leading to divergence.
 - A too-low learning rate can cause slow convergence, making training inefficient.
- Learning Rate Schedule:
 - Gradually decreasing the learning rate during training helps the model converge more smoothly and accurately.

- Types: Exponential decay, step decay, or adaptive learning rate methods like Adam.

b. Evaluate the role of activation functions in CNNs. How do they impact the learning ability of the network during training? (5)

- Activation Functions in CNNs:
 - They introduce non-linearity, allowing CNNs to learn complex patterns.
 - Common Activation Functions:
 - ReLU: Helps with gradient flow and sparsity.
 - Sigmoid/Tanh: Historically used but suffer from vanishing gradients.
- Impact: They ensure that the network can learn non-linear relationships and improve its capacity to generalize complex patterns.

5. a. Evaluate the impact of vanishing and exploding gradients in RNNs. How do these issues affect the network's ability to learn long-term patterns? (5)

- Vanishing Gradients:
 - Gradients become very small as they propagate backward, leading to slow or ineffective learning.
- Exploding Gradients:
 - Gradients become very large, causing weights to grow uncontrollably, destabilizing the network.
- Impact on Learning Long-Term Patterns:
 - Both issues hinder RNNs from capturing long-term dependencies. This is because the gradients needed to adjust weights for distant time steps either vanish or explode.

b. Evaluate the significance of the input, forget, and output gates in LSTMs. How do they contribute to the network's ability to remember long-term dependencies? (5)

- LSTM Gates:
 - Input Gate: Controls how much new information is added to the cell state.

- Forget Gate: Decides how much of the previous cell state should be forgotten.
- Output Gate: Determines how much of the current cell state should be passed to the output.
- Impact on Long-Term Dependencies:
 - These gates allow LSTMs to regulate memory more effectively, addressing the vanishing gradient problem and enabling them to capture long-term dependencies.